

# Executive Proposal: Enterprise Data & Infrastructure Modernisation

**Document Version:** 2.0

**Author:** Lead Systems Architect & AI Infrastructure Working Group

**Target Audience:** IT Management, Executive Steering Committee & Enterprise Architects

**Infrastructure Scope:** aws-3tier-deployment-for-php-infra & bda-ai-infra

## Executive Summary & Vision

This proposal outlines the comprehensive, end-to-end modernisation roadmap for our enterprise IT data and cloud infrastructure. It presents the historical evolution, current cloud-native baseline, and strategic future-proof target state for the organisation's core applications and Big Data Analytics & AI Infrastructure ( bda-ai-infra ).

We are transitioning from legacy monolithic virtual machines and static, proprietary visualization tools (Tableau) to an **API-First, Open-Source, and Model Context Protocol (MCP)-Ready** ecosystem running on AWS Graviton ( ap-southeast-5 ) and containerised rootless Podman pods.

| EVOLUTION OF IT INFRASTRUCTURE        |                                           |                                                 |
|---------------------------------------|-------------------------------------------|-------------------------------------------------|
| Historical State<br>(Legacy Monolith) | Current Baseline<br>(AWS 3-Tier OpenTofu) | Future-Proof Target<br>(API & MCP AI Ecosystem) |
| Single-node VM                        | Multi-AZ OpenTofu IaC                     | Rootless Podman Pods                            |
| Monolithic PHP                        | CodeIgniter 4 + PHP-FPM                   | Microservices + Fusio                           |
| Tableau Reports                       | AWS RDS Multi-AZ                          | Patroni PG18 + pgvector                         |
| Manual Ops                            | ASIMP, Wazuh & CloudWatch                 | Native LLM & MCP Tools                          |

# 1. Comprehensive Historical Context & Operational Evolution

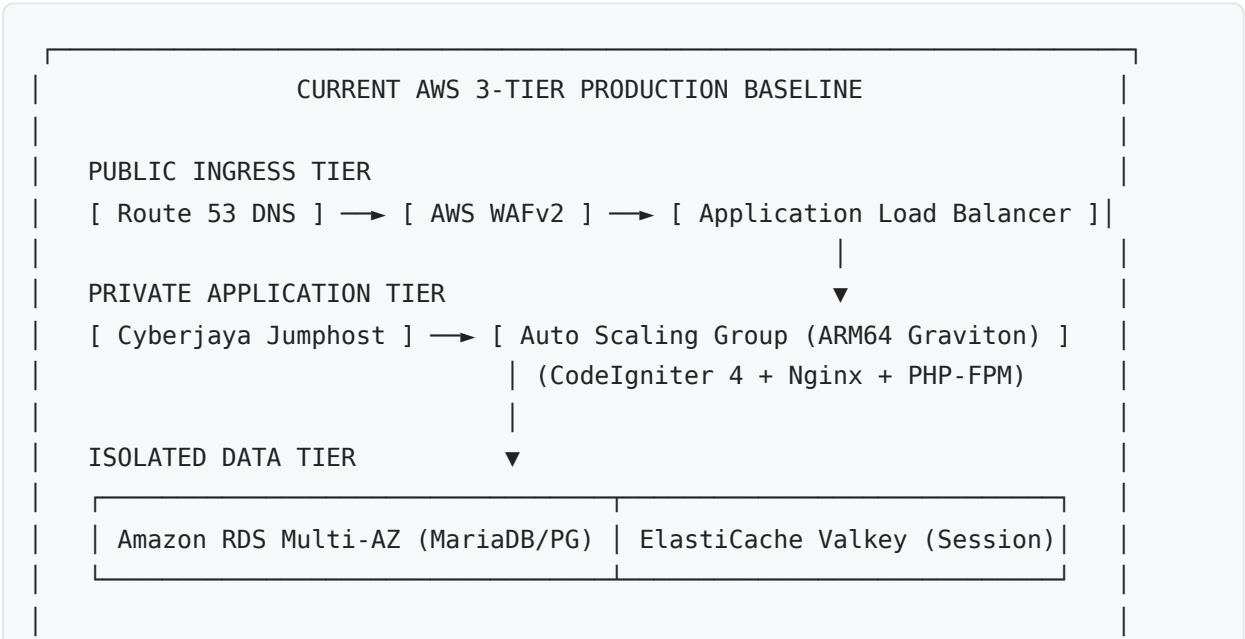
## 1.1 Phase 0: Historical Legacy Architecture

Historically, operations relied on standalone virtual machines running monolithic web applications alongside local database instances and proprietary BI dashboards (Tableau).

- **Operational Bottlenecks:** Single points of failure (SPOF), manual operating system upgrades, unencrypted local session storage, and lack of automated disaster recovery.
- **Licensing Liabilities:** High recurring per-user and per-core Tableau licensing fees, locking critical business calculations inside binary `.twb / .twbx` workbooks.
- **Data Silos:** Reporting was unidirectional. External systems, IoT sensors, and automated pipelines could not interact programmatically with enterprise business data without manual exports.

## 1.2 Phase 1: Current Production Baseline (AWS 3-Tier OpenTofu Stack)

To eliminate legacy single-node risks, the engineering team successfully migrated core web workloads to a hardened, cloud-native **AWS 3-Tier Architecture in the Asia Pacific (Malaysia) region ( `ap-southeast-5` )**.



Key accomplishments of the current baseline include:

1. **Infrastructure as Code (IaC):** Fully modular OpenTofu/Terraform code ( `vpc` , `alb` , `asg` , `rds` , `elasticache` , `waf` , `jumphost` , `fusio` , `standalone_ec2` ).
2. **Zero-Trust Network Isolation:** Public ALB ingress, isolated private application subnets, and database engines in non-routable private subnets.
3. **Security & Telemetry Consolidation:** Integration of ASIMP (Ansible System Integrity Management Platform), Lynis, OpenSCAP CIS Level 2 audits, Wazuh SIEM/XDR, and Amazon CloudWatch APM/RUM observability.
4. **Disaster Recovery (DR) Readiness:** Evaluated 3+1 DR strategies with regional same-account and cross-account ( `ap-southeast-5` ) isolation playbooks under Malaysian PDPA Section 129 compliance.

## 2. Core Strategic Pillars: API-Ready & MCP-Ready AI Modernisation

Building upon our current AWS 3-Tier baseline, the next strategic phase modernises the enterprise data layer into an **API-Ready and Model Context Protocol (MCP)-Ready** Big Data Analytics & AI Infrastructure ( `bda-ai-infra` ).

flowchart TD

```
subgraph Ingestion ["Bidirectional Ingestion & Quarantine Layer"]
```

```
Ext["External Systems / IoT / Partner APIs / User Feeds"]
```

```
Laravel["Laravel Web Portal (Human Verification Gate)"]
```

```
end
```

```
subgraph Core ["Central Data & Vector Engine (Tier 0 Golden SSoT)"]
```

```
NiFi["Apache NiFi 2.0 (Single Authoritative DB Writer)"]
```

```
Postgres["Percona Patroni PostgreSQL 18 + pgvector\n(Tier 0 Golden SSoT & Vector Store)"]
```

```
subgraph Integration ["Integration & API Layer (Fine-Grained Access Control)"]
```

```
REST["Open REST / gRPC APIs\n(Fusio API Server - Port 8080/443)"]
```

```
MCP["MCP Server Gateway\n(Model Context Protocol - Port 8443)"]
```

```
end
```

```

subgraph Consumers ["Consumers & Autonomous AI Systems"]
  BI["Open BI (Apache Superset / Metabase)"]
  AI["LLMs & Autonomous AI Agents\n(Google Jules & Antigravity)"]
end

Ext -->|"Ingress Payload"| Laravel
Laravel -->|"Verified Human Sign-off"| NiFi
NiFi -->|"Cryptographic Persistence"| Postgres
Postgres -->|"FGAC SQL Views"| REST
Postgres -->|"FGAC Vector & Tool Context"| MCP
REST -->|"Programmatic Inquiries"| BI
MCP -->|"Native Tool Execution"| AI

```

## 2.1 API-Ready: Universal Data Exchange

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- **Bidirectional Data Flow:** Enterprise applications, IoT edge devices, and partner systems inject structured telemetry directly via REST (Fusio API) or gRPC endpoints, as well as extract real-time operational streams.
- **Zero-Friction Interoperability:** Replaces proprietary connectors with standard JSON/REST interfaces accessible across web apps, mobile clients, and backend services.

## 2.2 MCP-Ready: Native AI & LLM Integration

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- **Direct AI Agent Integration:** Implements the open standard **Model Context Protocol (MCP)**. Large Language Models (LLMs) and autonomous agents (such as Google Jules and Antigravity) directly query database metrics, trigger background transformations, and retrieve vector embeddings as native "Tools".
- **Contextual Grounding:** Replaces manual static PDF/Excel exports with conversational, context-aware AI interactions connected directly to live database state.

## 2.3 Fine-Grained Access Control (FGAC) & Data Governance

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- **Row & Column Level Security:** Permissions are strictly enforced at the API gateway and database layer using session context injection ( `SET LOCAL` ). External applications or AI agents access only the precise data slices authorised for their authenticated identity.
- **Human SSoT & AI Provenance Tagging ( `bda_provenance` ):** Human-generated data remains the sole authoritative **Single Source of Truth (SSoT)** validated

through the human-in-the-loop verification gate. Any dataset created, modified, or enriched by AI processes is explicitly tagged using `bda_provenance` metadata.

```
{
  "bda_provenance": {
    "signature": "e2a9b418...",
    "key_id": "key-ed25519-prod-01",
    "verification_status": "VERIFIED_AI_ENRICHED",
    "verification_timestamp": "2026-08-11T12:00:00+08:00",
    "signature_algorithm": "Ed25519",
    "signature_encoding": "HEX_RAW_64_BYTE"
  }
}
```

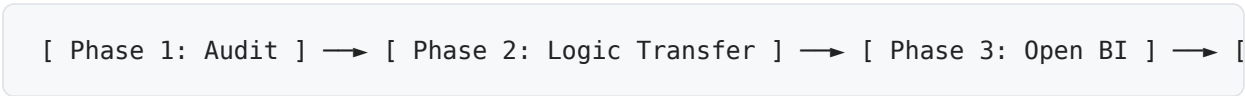
### 3. Summary Routing & Data Ingress Table

| Source Component            | Target Component            | Protocol / Port      | Ingress Security Boundary               | Operational Significance & Flow                                                                    |
|-----------------------------|-----------------------------|----------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------|
| External Systems / Users    | Laravel Verification Gate   | HTTPS (443) / REST   | Keycloak OAuth2 / Session Auth          | Ingests non-IT user feeds and telemetry into quarantine staging directory.                         |
| Laravel Verification Gate   | Apache NiFi 2.0 Ingest Gate | Internal Event Spool | Human Signature / Audit Log             | Triggers approval workflow upon explicit human verification.                                       |
| Apache NiFi 2.0 Ingest Gate | PostgreSQL 18 SSoT Store    | TCP 5432 / Native PG | <code>nifi_ingest_writer</code> DB Role | Sole authoritative writer committing Tier 0 Golden SSoT data and <code>bda_provenance</code> tags. |

| Source Component         | Target Component          | Protocol / Port          | Ingress Security Boundary      | Operational Significance & Flow                                                          |
|--------------------------|---------------------------|--------------------------|--------------------------------|------------------------------------------------------------------------------------------|
| PostgreSQL 18 SSoT Store | Fusio REST / gRPC Gateway | TCP 5432 / Read Views    | PostgreSQL Row-Level Security  | Exposes fine-grained SQL views and endpoints to web apps and BI dashboards.              |
| PostgreSQL 18 SSoT Store | MCP Tool Gateway          | TCP 5432 / Vector Search | MCP Tool Scope & Ed25519 Token | Serves vector context and structured database tool capabilities to autonomous AI agents. |

## 4. Tableau Migration & Modernisation Strategy

Legacy Tableau workbooks will be systematically decommissioned using a four-phase migration roadmap to ensure zero downtime and manage operational risk:



- Phase 1: Workbook Audit & Inventory:** Catalogue all active Tableau workbooks ( .twb / .twbx ), calculated fields, custom SQL queries, and user permission matrices. Identify redundant reports and mark high-value dashboards for migration.
- Phase 2: Data & Logic Consolidation:** Migrate complex Tableau calculations and data blending rules directly into PostgreSQL Materialised Views and stored functions. Ensure Apache NiFi orchestrates clean data pipelines into normalized schemas.
- Phase 3: Open-Source BI Deployment:** Deploy containerised Apache Superset (or Metabase) on Podman to replicate executive dashboards with zero user-license fees.
- Phase 4: API & MCP Enablement:** Expose business metrics as REST/gRPC API endpoints and wrap vector searches into standardised MCP Tools for internal AI agent workflows.

## 5. Podman Infrastructure & Container Blueprint

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The target infrastructure relies on rootless Podman pods to enforce high availability, zero vendor lock-in, and full Kubernetes compatibility.

```
# Conceptual Architecture Blueprint: podman-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: bda-ai-infra-pod
spec:
  containers:
    - name: postgres-engine
      image: docker.io/pgvector/pgvector:pg18
      description: "Central database store equipped with vector search capabilities"

    - name: nifi-orchestrator
      image: docker.io/apache/nifi:latest
      description: "Low-latency data ingestion, batch processing, and ETL orchestration"

    - name: mcp-api-gateway
      image: localhost/bda-mcp-server:v1
      description: "Custom Python/Node.js gateway delivering Open APIs, MCP Tools, and AI Agents"

    - name: open-bi-superset
      image: docker.io/apache/superset:latest
      description: "Open-source business intelligence platform replacing Tableau."
```

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## 6. Financial & Operational ROI (3-Year TCO Comparison)

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To evaluate Total Cost of Ownership (TCO) in the Malaysia region ( `ap-southeast-5` ), we compare the legacy setup, self-hosted EC2 stacks, and the proposed API/MCP Podman stack over a 36-month timeline (1 USD  $\approx$  4.50 MYR).

| Area               | Legacy Architecture (Tableau-based)                    | Self-Hosted Custom Stack (EC2)                 | Proposed AWS + Podman Stack                                            |
|--------------------|--------------------------------------------------------|------------------------------------------------|------------------------------------------------------------------------|
| Licensing Costs    | High recurring per-user (\$42/user/mo) and core fees.  | \$0.00 proprietary software fees.              | <b>\$0.00 proprietary software fees (100% Open Source).</b>            |
| Data Accessibility | Locked inside proprietary <code>.twb</code> workbooks. | Direct SQL queries requiring DB access.        | <b>Universal API &amp; MCP endpoints accessible by any tool/agent.</b> |
| Engineering OpEx   | High manual maintenance & export labor.                | High (\$1,500/mo DBRE labor for Patroni/etcd). | <b>Optimised (\$150/mo automated OpenTofu management).</b>             |
| AI Integration     | None (Manual Excel exports required).                  | Custom custom-built agent scripts.             | <b>Native MCP support for autonomous AI workflows.</b>                 |
| 3-Year Total TCO   | ~\$110,000.00 USD (MYR 495,000)                        | \$78,016.68 USD (MYR 351,075)                  | <b>\$39,430.80 USD (MYR 177,438)</b>                                   |

## Financial Impact Summary

Transitioning to the proposed AWS + Podman API/MCP Architecture saves **\$38,585.88 USD (~RM 173,636.46 MYR)** over 36 months compared to self-managed EC2 stacks, and over **\$70,500.00 USD (~RM 317,250.00 MYR)** compared to legacy Tableau setups—representing a **49.5% to 64.1% net savings**.

## 7. Next Steps & Execution Roadmap

Upon approval of this proposal, execution will proceed as follows via automated code and configuration updates:

- Commit Proposal Document:** Save `docs/IT-MANAGEMENT-PROPOSAL.md` into the main repository branch.
- Deploy Podman Pod Spec:** Commit container definitions for PostgreSQL 18 pgvector, NiFi, and Superset in rootless Podman configurations.
- Build MCP Gateway Skeleton:** Deploy the Python/Node.js MCP gateway with PostgreSQL tool connections and FGAC middleware.

4. **Initiate Phase 1 Migration:** Begin Tableau workbook audit and SQL logic extraction.

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*Deep State of Mind (DSOM) For My AI Protocol Harisfazillah Jamel (LinuxMalaysia)  
2026-08-11 Standard: UK English DBP-standard Bahasa Melayu Malaysia (Piawai)  
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[ REL: 3.5.1 ] | [ STD: RFC\_9116 ] | [ ENV: OPENTOFU\_1.6 ] | [ VIEW: STANDARD ]

Rendered: Statically Compiled at Build-time | MEM: 0 KB (Zero-runtime database-free)